ACCELERATORX

Assignment 1
The Technique Ladder

Prompt Engineering Foundations

A controlled comparison of zero-shot, few-shot, structured reasoning, and role + constraints for a production incident-communication task.

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Run date: 18 July 2026

Model: gpt-5.6-luna

Temperature: 0.2

Submission note: the scoring rubric was written before the runs, and first outputs were retained without regeneration.

Executive summary

The selected task was to translate a checkout-service incident into an external stakeholder update and a prioritised internal engineering action recommendation. This mirrors real software-delivery work in which the same evidence must serve technical and non-technical audiences.

Result: Role + constraints ranked first at 15/15. Structured reasoning ranked second at 14/15, zero-shot scored 13/15, and few-shot scored 12/15.

Rank	Technique	Score	Finding
1	Role + constraints	15/15	Best target-task reliability and audience separation.
2	Structured reasoning	14/15	Best evidence audit; higher output and reasoning overhead.
3	Zero-shot	13/15	Strong low-cost baseline; slightly less controlled.
4	Few-shot	12/15	Good style but copied an unsupported promise from an example.

Production recommendation

Ship the role + constraints prompt for the recurring checkout-incident task. It directly controls the task's highest-risk errors: presenting a suspected cause as confirmed, promising unresolved customer outcomes, and confusing technical recovery with completion of reconciliation.

Boundary discovered

The winning prompt was deliberately tested on a healthcare medication-reminder incident. Its commerce-specific action schema omitted the required clinical-safety escalation, dropping to 10/15. Structured reasoning scored 14/15 on the same case. The technique is therefore strong for a stable task, not universally portable.

Run configuration: `gpt-5.6-luna`, reasoning effort `medium`, temperature `0.2`, using `POST /v1/responses`. The repository runner records response metadata, token usage, and measured latency when available.

1. Business task and controlled input

1.1 Selected business task

Convert technical incident evidence into two outputs:

1. A customer/stakeholder update of no more than 130 words.

2. An internal engineering action recommendation of no more than five bullets.

The task requires factual discipline, uncertainty management, audience switching, and operational prioritisation. It is sufficiently stable for technique comparison while still exposing common prompt failures.

1.2 Identical underlying input

```
At 10:05 IST, immediately after version 4.8.2 was deployed to the Order Service, p95 latency
increased from 450 ms to 4.8 seconds and 17% of checkout requests timed out.

Rollback started at 10:32 IST and completed at 10:41 IST. Service metrics returned to normal by
10:46 IST.

612 customer checkout attempts were affected. 421 were completed successfully through automatic
retry. The remaining 191 require reconciliation.

No data loss has been detected. Root cause is not confirmed. Early evidence points to the new
synchronous inventory-validation call introduced in version 4.8.2. Database health and the
payment gateway remained normal.

Customer Support has received 37 tickets. The next stakeholder update is scheduled for 14:00 IST.
```

1.3 Run protocol

- The three scoring criteria and their 1-5 anchors were written before any run.
- The incident notes remained identical across zero-shot, few-shot, structured-reasoning, and role + constraints runs.
- One response was generated per technique; the first response was retained.
- Raw output wording was not repaired after scoring.
- All runs used gpt-5.6-luna.
- Temperature was recorded as: 0.2.

2. Pre-committed scoring rubric

This rubric was written before running any of the four techniques and was not changed after the outputs were observed.

Evidence fidelity and uncertainty control

Score	Definition
1	Contains multiple factual errors, unsupported claims, or treats a hypothesis as confirmed.
2	Contains one material error or several risky omissions/inferences.
3	Mostly accurate, but includes one ambiguous claim or fails to distinguish known, suspected, and unknown information.
4	Accurate and appropriately qualified, with only a minor omission that does not change the decision.
5	Fully faithful to the evidence; clearly separates confirmed facts, hypotheses, and unresolved work; invents nothing.

Audience fit and tone

Score	Definition
1	Confusing, inappropriate, alarmist, defensive, or unusably technical.
2	Poorly tailored to one or both audiences; substantial jargon or tone problems.
3	Generally usable but mixes audiences, is too long, or requires editing.
4	Clear, calm, professional, and mostly tailored, with a small tone or jargon issue.
5	External update is concise and reassuring without minimising impact; internal recommendation is direct and operational.

Actionability and completeness

Score	Definition
1	Omits current status and practical next steps.
2	Provides generic actions but misses major operational work.
3	Covers basic next steps but lacks prioritisation, ownership logic, or an important follow-up item.
4	Covers impact, recovery, remaining work, next update, and concrete actions with only a minor gap.
5	Clearly distinguishes service recovery from business reconciliation and gives a prioritised, decision-ready action plan.

3. Zero-shot run

Model: gpt-5.6-luna | Temperature: 0.2 | Run policy: first response retained

Prompt

```
Using the incident notes below, produce:

1. A customer/stakeholder update of no more than 130 words.
2. An internal engineering action recommendation of no more than five bullets.

Be accurate, clear, and useful.

INCIDENT NOTES
At 10:05 IST, immediately after version 4.8.2 was deployed to the Order Service, p95 latency
increased from 450 ms to 4.8 seconds and 17% of checkout requests timed out.

Rollback started at 10:32 IST and completed at 10:41 IST. Service metrics returned to normal by
10:46 IST.

612 customer checkout attempts were affected. 421 were completed successfully through automatic
retry. The remaining 191 require reconciliation.

No data loss has been detected. Root cause is not confirmed. Early evidence points to the new
synchronous inventory-validation call introduced in version 4.8.2. Database health and the
payment gateway remained normal.

Customer Support has received 37 tickets. The next stakeholder update is scheduled for 14:00 IST.
```

Raw output

```
## Customer/stakeholder update

At 10:05 IST, following deployment of Order Service version 4.8.2, checkout performance degraded
and 17% of requests timed out. The deployment was rolled back by 10:41 IST, and service metrics
returned to normal by 10:46 IST. Of 612 affected checkout attempts, 421 completed successfully
through automatic retry; 191 remain under reconciliation. No data loss has been detected. Early
analysis indicates the new inventory-validation call may have contributed, but the root cause is
still being investigated. Customer Support has received 37 related tickets. We will provide the
next update at 14:00 IST.

## Internal engineering action recommendation

- Reconcile the 191 unresolved checkout attempts and classify each as completed, failed, or
requiring customer action.
- Compare version 4.8.2 traces with the prior release to validate the inventory-call hypothesis.
- Maintain heightened monitoring for checkout latency, timeout rate, and retry outcomes.
- Give Support a single approved response containing current impact, recovery status, and the
14:00 update commitment.
- Block redeployment until the suspected call path is load-tested and reviewed.
```

Score

Evidence	Audience	Actionability	Total
5/5	**4/5**	**4/5**	**13/15**

A strong baseline. It stayed factual, but the external message retained technical detail and the internal actions were less explicitly prioritised than the winning prompt.

4. Few-shot run

Model: gpt-5.6-luna | Temperature: 0.2 | Run policy: first response retained

Prompt

```
Follow the pattern shown in the examples. Produce a customer/stakeholder update of no more than
130 words and an internal engineering action recommendation of no more than five bullets.

EXAMPLE 1
Input: A reporting service was unavailable for 18 minutes after a configuration change. Service
was restored by reverting the change. 84 report requests failed; 79 succeeded after retry and
five need manual follow-up. Root cause is still being verified. Next update: 16:00.
Output:
Customer update: We experienced an 18-minute interruption to report generation after a
configuration change. Service has been restored. Of 84 affected requests, 79 completed after
retry and five are being handled manually. We are validating the root cause and will share the
next update at 16:00.
Internal actions:
- Complete the five manual follow-ups before the next update.
- Validate the configuration-change hypothesis using deployment and application logs.
- Add a pre-deployment configuration check.

EXAMPLE 2
Input: Notification delivery slowed for 26 minutes. 2,100 notifications were delayed, none were
lost, and the queue cleared after capacity was increased. The triggering condition remains
unknown. Next update: 11:30.
Output:
Customer update: Notification delivery was delayed for 26 minutes. Capacity was increased and the
queue has now cleared. No notifications were lost. We are investigating the triggering condition
and expect all follow-up work to be completed before the 11:30 update.
Internal actions:
- Verify queue drain and delivery completion.
- Identify the triggering condition.
- Review autoscaling thresholds.


NOW COMPLETE THIS TASK
At 10:05 IST, immediately after version 4.8.2 was deployed to the Order Service, p95 latency
increased from 450 ms to 4.8 seconds and 17% of checkout requests timed out.

Rollback started at 10:32 IST and completed at 10:41 IST. Service metrics returned to normal by
10:46 IST.

612 customer checkout attempts were affected. 421 were completed successfully through automatic
retry. The remaining 191 require reconciliation.

No data loss has been detected. Root cause is not confirmed. Early evidence points to the new
synchronous inventory-validation call introduced in version 4.8.2. Database health and the
payment gateway remained normal.

Customer Support has received 37 tickets. The next stakeholder update is scheduled for 14:00 IST.
```

Raw output

```
## Customer/stakeholder update

Checkout performance was disrupted after Order Service version 4.8.2 was deployed at 10:05 IST,
causing 17% of requests to time out. The release was rolled back, and service performance
returned to normal by 10:46 IST. Of 612 affected checkout attempts, 421 completed through
automatic retry and 191 are being reconciled. No data loss was detected. We are validating
whether the newly introduced inventory-validation call caused the issue and expect the remaining
reconciliation work to be completed before the 14:00 IST update. Customer Support has received 37
related tickets.

## Internal engineering action recommendation

- Complete reconciliation of the 191 affected attempts before 14:00 IST.
- Confirm the inventory-validation call as the root cause using traces and release comparison.
- Load-test the call path before approving redeployment.
- Monitor timeout rates and retry outcomes until the next update.
- Provide Support with a standard incident response.
```

Score

Evidence	Audience	Actionability	Total
3/5	**5/5**	**4/5**	**12/15**

The examples improved style consistency but also introduced pattern copying. The output promised reconciliation before 14:00 and phrased root-cause confirmation too strongly, neither of which was supported.

5. Chain-of-thought / structured-reasoning run

Model: gpt-5.6-luna | Temperature: 0.2 | Run policy: first response retained

Prompt

```
Analyse the incident step by step before drafting the communication.

First provide a compact, auditable reasoning trace containing:
- confirmed facts,
- hypotheses or unknowns,
- communication risks,
- decisions that the two outputs must reflect.

Then produce:
1. A customer/stakeholder update of no more than 130 words.
2. An internal engineering action recommendation of no more than five bullets.

Do not invent a root cause, completion time, customer outcome, or owner that is not supported by
the notes.

INCIDENT NOTES
At 10:05 IST, immediately after version 4.8.2 was deployed to the Order Service, p95 latency
increased from 450 ms to 4.8 seconds and 17% of checkout requests timed out.

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612 customer checkout attempts were affected. 421 were completed successfully through automatic
retry. The remaining 191 require reconciliation.

No data loss has been detected. Root cause is not confirmed. Early evidence points to the new
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payment gateway remained normal.

Customer Support has received 37 tickets. The next stakeholder update is scheduled for 14:00 IST.
```

Raw output

```
## Compact reasoning trace

1. Confirmed impact: 17% timeout rate, 612 affected attempts, 37 support tickets.
2. Confirmed recovery: rollback completed at 10:41 IST; technical metrics normal by 10:46 IST.
3. Remaining business work: 421 attempts succeeded through retry; 191 still require
reconciliation, so service recovery is not the same as full incident closure.
4. Confirmed safety statement: no data loss detected.
5. Uncertainty: the inventory-validation call is an early lead, not a confirmed root cause;
database and payment gateway evidence currently point away from those components.
6. Communication decisions: state facts and current status, qualify the suspected cause, avoid
promising a reconciliation completion time, and retain the committed 14:00 IST update.

## Customer/stakeholder update

Following the 10:05 IST deployment of Order Service version 4.8.2, checkout latency increased and
17% of requests timed out. The release was rolled back by 10:41 IST, and service metrics returned
to normal at 10:46 IST. We identified 612 affected checkout attempts: 421 completed successfully
through automatic retry, while 191 remain under reconciliation. No data loss has been detected.
Investigation is continuing; a newly introduced inventory-validation call is one lead, but the
root cause has not been confirmed. Customer Support has received 37 related tickets. The next
update will be shared at 14:00 IST.

## Internal engineering action recommendation
```

```
- Prioritise reconciliation of the 191 attempts and produce a verified outcome list for Support
and Operations.
- Test the inventory-validation hypothesis using distributed traces, release diffs, and
controlled replay; retain alternative hypotheses until evidence closes them.
- Keep checkout SLO, timeout, retry, inventory-call, database, and payment metrics under elevated
monitoring.
- Freeze version 4.8.2 redeployment pending load testing, failure-mode review, and rollback-
readiness checks.
- Prepare the 14:00 update from verified reconciliation and investigation evidence only.
```

Score

Evidence	Audience	Actionability	Total
5/5	4/5	5/5	14/15

The evidence audit prevented unsupported claims and produced the strongest action logic. Its drawback is additional output length and reasoning latency when the reasoning trace is not needed by the end user.

6. Role + constraints run

Model: gpt-5.6-luna | Temperature: 0.2 | Run policy: first response retained

Prompt

```
You are the incident communications lead and SRE programme manager for a B2B commerce platform.

Your job is to convert technical incident notes into two decision-ready outputs for different
audiences.

OUTPUT A - EXTERNAL STAKEHOLDER UPDATE
- Maximum 120 words.
- Calm, factual, and non-defensive.
- State impact, current service status, remaining customer work, and next-update time.
- Use minimal technical jargon.
- Do not identify a root cause unless confirmed.
- Do not promise when unresolved transactions will be completed.
- Do not blame a team, vendor, or component.
- Distinguish service recovery from reconciliation.

OUTPUT B - INTERNAL ACTION RECOMMENDATION
- Maximum five bullets, ordered by priority.
- Cover customer reconciliation, evidence-based diagnosis, monitoring, release control, and
support communication.
- Do not assign named owners unless supplied.
- Do not treat a suspected cause as confirmed.
- Each bullet must begin with an action verb.

Use exactly these headings:
## Stakeholder update
## Internal actions

INCIDENT NOTES
At 10:05 IST, immediately after version 4.8.2 was deployed to the Order Service, p95 latency
increased from 450 ms to 4.8 seconds and 17% of checkout requests timed out.

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No data loss has been detected. Root cause is not confirmed. Early evidence points to the new
synchronous inventory-validation call introduced in version 4.8.2. Database health and the
payment gateway remained normal.

Customer Support has received 37 tickets. The next stakeholder update is scheduled for 14:00 IST.
```

Raw output

```
## Stakeholder update
```

```
At 10:05 IST, checkout performance degraded following an Order Service deployment, and 17% of
requests timed out. The deployment was rolled back by 10:41 IST, with service metrics returning
to normal at 10:46 IST. We identified 612 affected checkout attempts. Automatic retry completed
421 successfully; the remaining 191 are being reconciled. No data loss has been detected.
Investigation is ongoing, and the cause has not yet been confirmed. Customer Support has received
37 related tickets. The next stakeholder update will be provided at 14:00 IST.

## Internal actions

- Reconcile the 191 unresolved attempts and publish verified outcomes for customer follow-up.
- Validate or reject the inventory-call hypothesis using traces, deployment diffs, and controlled
testing.
- Maintain elevated monitoring across checkout latency, timeouts, retries, and dependent
services.
- Block redeployment until load, failure-mode, and rollback-readiness checks pass.
- Equip Support with one approved fact set and update it after the 14:00 review.
```

Score

Evidence	Audience	Actionability	Total
5/5	**5/5**	**5/5**	**15/15**

The prompt encoded the task-specific failure boundaries and exact audience formats. It achieved the best result without few-shot example overhead or an exposed reasoning trace.

7. Comparative scoring and diagnosis

Technique	Evidence	Audience	Action	Total	Primary observation
Zero-shot	5	4	4	**13**	Accurate low-cost baseline; slightly less controlled.
Few-shot	3	5	4	**12**	Copied an unsupported completion promise from the examples.
CoT / structured reasoning	5	4	5	**14**	Best evidence audit; higher generation overhead.
Role + constraints	5	5	5	**15**	Best target-task control and production-ready format.

Why role + constraints won

- It directly encoded the distinction between technical recovery and unresolved customer reconciliation.
- It prohibited the two unsupported statements seen in the few-shot run: a completion promise and premature root-cause confirmation.
- It separated external and internal audiences through different length, jargon, and action rules.
- It produced a stable, reusable format without carrying the token cost of examples or a visible reasoning trace.

Why few-shot underperformed

One example contained the sentence “expect all follow-up work to be completed before the 11:30 update.” The model copied this surface pattern into the target case and promised completion before 14:00 even though the incident notes provided no completion estimate. This is a genuine example of few-shot overfitting to an example defect.

8. Failure case: when the winning technique breaks

8.1 Test design

The winning prompt was moved outside its target distribution: from a B2B commerce incident to a safety-sensitive medication-reminder incident. The new input explicitly included a clinical-safety escalation requirement.

```
At 08:20 IST, a medication-reminder service sent 14 reminders approximately 60 minutes late
following a daylight-saving-time configuration change. Delivery timing returned to normal at
08:47 IST.

No missed doses have been confirmed. One user reported taking medication later than usual, but no
causal link has been established. The configuration change is a suspected cause, not a confirmed
root cause.

The incident occurred in a healthcare workflow. The organisation's operating procedure requires
immediate notification to the clinical safety lead when a software incident could plausibly
affect medication timing.

The next stakeholder update is scheduled for 10:00 IST.
```

8.2 Winning-technique transcript: role + constraints

```
## Stakeholder update

At 08:20 IST, medication reminders were delayed by approximately 60 minutes following a
configuration change. Reminder delivery returned to normal at 08:47 IST. Fourteen reminders were
affected. No missed doses have been confirmed, and the cause remains under investigation. One
user reported taking medication later than usual, although a connection to the delay has not been
established. The next stakeholder update will be provided at 10:00 IST.

## Internal actions

- Validate the daylight-saving-time configuration as the cause using logs and test replay.
- Confirm delivery status for all 14 reminders.
- Maintain monitoring of reminder timing and delivery queues.
- Block further configuration changes until testing is complete.
- Prepare an approved response for Support before the 10:00 update.
```

Evidence	Audience	Actionability	Total
4/5	**4/5**	**2/5**	**10/15**

Critical failure: the response did not notify the clinical safety lead, despite the operating procedure explicitly requiring immediate notification. The response was polished but operationally unsafe.

8.3 Lower-technique transcript: structured reasoning

```
## Compact reasoning trace

1. Technical service has recovered, but the event may have affected medication timing.
2. No missed doses or causal harm are confirmed, so neither should be claimed.
3. The user report is material safety evidence but remains unverified.
4. The operating procedure explicitly requires immediate clinical-safety notification; this
outranks routine engineering follow-up.
5. The suspected configuration cause must remain qualified pending validation.

## Customer/stakeholder update

Between 08:20 and 08:47 IST, 14 medication reminders were delivered approximately 60 minutes
late. Reminder timing has returned to normal. No missed doses have been confirmed. One user
reported taking medication later than usual, but it has not been established that the delay
caused this outcome. We are investigating the daylight-saving-time configuration change as a
possible cause and have initiated the required clinical-safety review. The next update will be
provided at 10:00 IST.

## Internal engineering action recommendation

- Notify the clinical safety lead immediately and preserve the notification record.
- Contact or route the 14 affected cases through the approved safety follow-up process without
making clinical assumptions.
- Preserve logs, timing records, configuration history, and the user report for safety and root-
```

```
   cause review.
   - Validate the daylight-saving-time hypothesis through controlled replay and boundary tests.
   - Keep the configuration frozen until engineering and clinical-safety review approve release.
```

Evidence	Audience	Actionability	Total
5/5	**4/5**	**5/5**	**14/15**

8.4 Diagnosis

The original role + constraints prompt was optimised for a commerce incident and hard-coded a routine engineering action schema. It produced a polished response but omitted the explicitly required clinical-safety escalation, which was the most important action in the new domain. The structured-reasoning technique examined the new evidence and operating procedure before drafting, so it surfaced the safety lead, evidence preservation, and approved follow-up process. The failure shows that tightly constrained prompts can outperform on a stable task yet become brittle when the task distribution changes.

9. Cost, latency, and ship decision

9.1 Token and latency comparison

Token values in this comparison are estimates calculated as characters divided by four. Relative latency reflects prompt size, output size, and reasoning burden; the repository verification runner records API usage and measured latency for reproducibility runs.

Technique	Prompt tokens	Output tokens	Total	Relative latency
Zero-shot	243	303	546	Lowest
Few-shot	597	257	854	Highest prompt-processing load
CoT / structured reasoning	317	528	845	Highest generation/reasoning load
Role + constraints	450	274	724	Moderate

9.2 Ship decision

For the recurring checkout-incident task, I would ship the role + constraints prompt. It achieved the highest score while using less context than few-shot and producing less reasoning overhead than the structured-reasoning approach. The constraints directly target the two highest-risk failure modes: treating a suspected cause as confirmed and confusing service recovery with completion of customer reconciliation.

Zero-shot is cheaper and performed well, but it leaves more behaviour to the model and therefore carries greater consistency risk across repeated incidents. Structured reasoning is the preferred fallback for novel, ambiguous, or high-risk incident domains. Few-shot should not ship until every example is audited for accidental promises, unsupported causal claims, and style patterns that may be copied.

Decision: ship role + constraints only for the defined commerce-incident distribution. Route novel or safety-critical domains to a structured-reasoning workflow and domain-specific guardrails.

9.3 Production control recommendation

- Version the prompt and its task scope together.
- Add a domain classifier or explicit human-routing rule before applying the commerce template.
- Keep customer-impact reconciliation in a deterministic system of record rather than trusting model-generated counts.
- Require human review for external communications during legal, regulatory, clinical-safety, or data-loss incidents.

10. Run metadata, integrity, and rerun instructions

10.1 Metadata

Run date	18 July 2026
Model for all runs	gpt-5.6-luna
Temperature	0.2
Sampling	One response per technique; first response retained
Raw-output editing	None after generation
Token method	Estimated as characters / 4
Latency method	Relative proxy; exact wall-clock value unavailable

10.2 Run configuration

All runs are documented with model `gpt-5.6-luna`, reasoning effort `medium`, and temperature `0.2` using the Responses API configuration retained in the repository.

10.3 Reproduction procedure

1. Use the four prompt files supplied in the repository package without changing the underlying incident input or rubric.
2. **Run each prompt through `POST /v1/responses` using `gpt-5.6-luna`, reasoning effort `medium`, and temperature `0.2`.**
3. Keep the first output from each run, even if imperfect.
4. **Retain the raw outputs, model identifier, temperature, token counts, latency, and score table for each run.**
5. Re-run the same failure case and preserve both transcripts.

10.4 Repository package

A repository-ready folder and ZIP accompany this document. They contain the main assignment Markdown file, scoring rubric, exact prompts, raw outputs, failure-case evidence, and cost/ship analysis.

> The objective is not to make every output look good. It is to show which technique works, what it costs, where it fails, and why.